SECTION 3
CRATED MAGNET DELIVERY

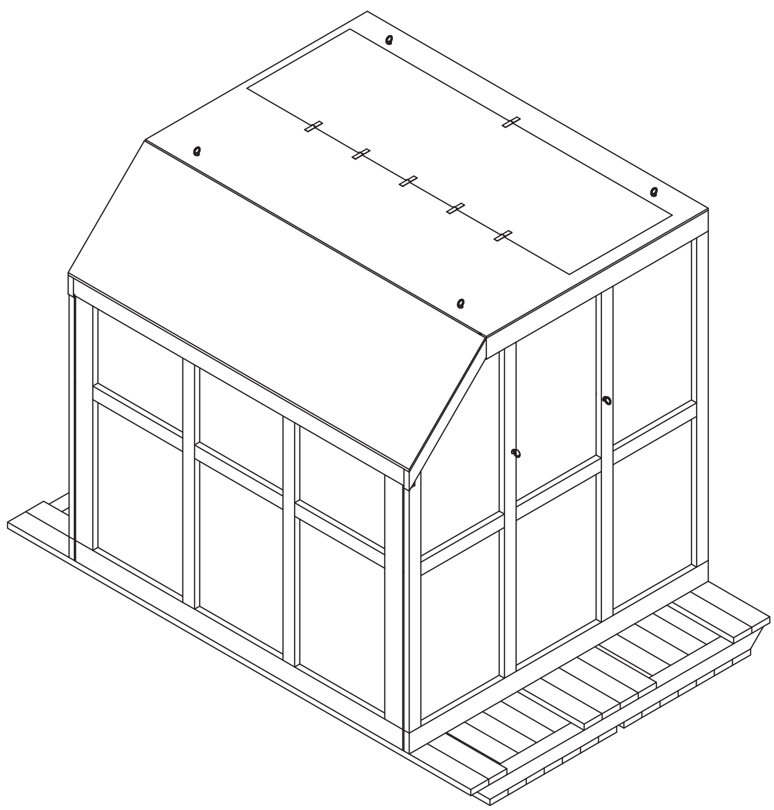
WARNING!

LIFTING / JACKING ANGLES THAT ARE SHIPPED INSTALLED ON MAGNET ARE NOT TO BE USED TO LIFT MAGNET WITH CRANE. SERIOUS PERSONNAL INJURY AND MAGNET DAMAGE COULD RESULT. MAGNET MAY BE LIFTED USING BELLY STRAPS WITH CRANE OR LIFTING / JACKING ANGLES WITH FORKLIFT ONLY. REFER TO THIS MANUAL FOR PROPER LIFTING CONFIGURATIONS / REQUIREMENTS.

3–1 EQUIPMENT / TOOLS FOR CRATED MAGNET INSTALLATION

Note

The following information outlines provisions for filling and electrical / temperature checks while magnet is still in shipping crate. This will enable required service to be performed in transit based upon magnet shipping date.



ACCESS DOOR LABELING ON CRATE
ILLUSTRATION 3–1

3–1 EQUIPMENT / TOOLS FOR CRATED MAGNET INSTALLATION (continued)

3–1–1 Forklift / Crane

ITEM	QTY.	CAPACITY / RANGE	FURNISHED BY:	INTENDED USE:
Forklift	1	14,500 lbs. (6,577 Kg) at 50” (1,270 mm) to load center Minimum fork length 90” (2,286 mm) Minimum distance between forks 40” (1,016 mm)	Rigger	Unloading or moving magnet with pallet and crate installed
Forklift	1	12,000 lbs. (5,450 Kg) at 50” (1,270 mm) to load center Minimum fork length 90” (2,286 mm) Minimum distance between forks 70” (1,778 mm) Top lift minimum fork height 138” (3,505 mm)	Rigger	Unloading or moving magnet (no pallet)
Crane	1	12,000 lbs. (6,124 Kg)	Rigger	Unloading or moving magnet

3–1–2 Crane Lift Using Slings Around Belly of Magnet

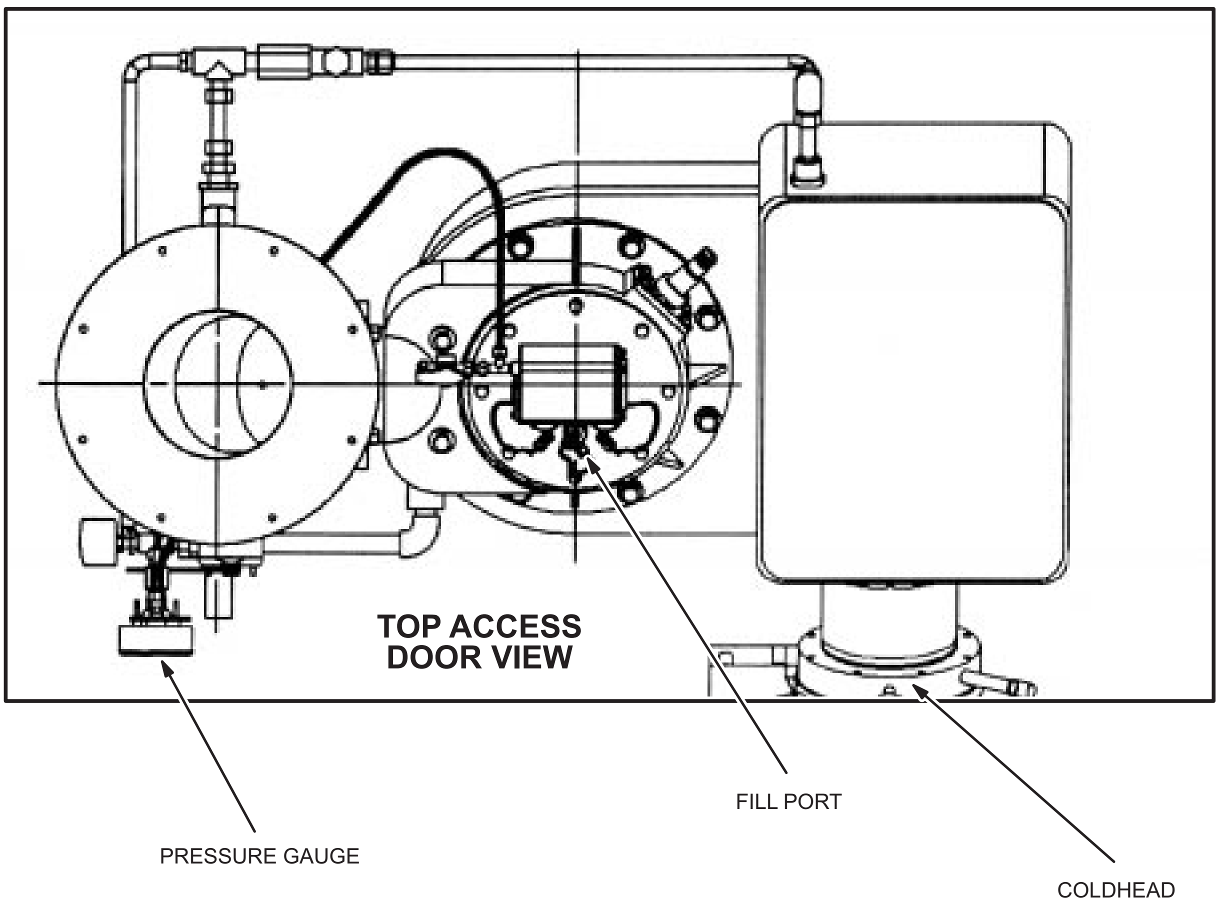
Spreader Beam	1	12,000 lbs. (6,124 Kg) 48” – 50” (1,219 mm – 1,270 mm) between lift points	Rigger	Unloading or moving magnet
Single Ended Strap	2	12,000 lbs. (6,124 Kg) Total weight for both straps combined. Minimum 24 ft. (7.32 M) long and 3” (76.2 mm) wide	Rigger	Unloading or moving magnet

3–1 EQUIPMENT / TOOLS FOR CRATED MAGNET INSTALLATION (continued)

3–1–3 Miscellaneous Equipment / Tools

Hydraulic or Toe Jack	2 or 4	Must support one end of magnet on two jacks or both ends of magnet on 4 jacks Total weight 12,000 lbs. (5,450 Kg) 7 1/2” (190 mm) from foot bottom to jacking surface.	Rigger	Raise up magnet for roller dollies or leveling plates
Roller Dollies	2, 3 or 4	12,000 lbs. (5,450 Kg) Total weight for all dollies combined	Rigger	Moving magnet to magnet room
Level	1	4’ (1,219 mm) minimum	Rigger	Level the magnet
Magnet Leveling Kit	1	2,500 lbs.	GE / SYS	46–260888G4 / Level the magnet
Magnet Boltdown / Seismic Kit	1	Apply sufficient torque to provide 12,000 lbs. (5,450 Kg) clamping force per anchor. Torque requirements to be supplied by Screen Room Vendor.	Screen Room Vendor	Bolt down magnet
Torque Wrench and Socket	1	Torque range (.625” bolt x 1.00” long)	Rigger	Bolt down magnet
Socket Wrench	1	3/4” socket size	Rigger	Remove 1/2” crate lag screws
Magnet Interface Drawing	1	N / A	GE / ISS	2183478IDW Identify magnet dimensions and features
Pre–delivery Information Package	1	N / A	GE	Aid in magnet installation
Portable Temperature Meter	1	N / A	GE / FE	2125073 / Check internal temperatures
Digital Volt / Ohm Meter	1	Measurement capability: < 1 mA	GE / FE	Perform magnet electrical checks
Magnet System Components Checks: Tables 3–1 to 3–5	1	N / A	GE / SYS	Confirm electrical check values

3–2 IN–TRANSIT HANDLING AND SERVICE



TOP ACCESS DOOR
ILLUSTRATION 3–2

3–2 IN–TRANSIT HANDLING AND SERVICE (continued)

CAUTION

Review guidelines with Shipper / Riggers prior to transporting magnet.

Shipping and handling guidelines are provided in Tables 3–1, 3–2 and 3–3.

These guidelines must be followed to prevent any potential damage to the magnet during shipping and handling.

LIFTING / HANDLING INFORMATION
TABLE 3–1

MAGNET	* MAXIMUM WEIGHT	** MAXIMUM TILT	FORKLIFT CAPABILITY
LCC 1.5T	12,000 lbs. (5,450 kg.)	30 deg.	Yes
LCC 1.0T			

CAUTION

The magnet CANNOT be shipped by train as serious damage to the magnet's internal suspension system may occur due to the vibration loads introduced by rail systems.

SHIPPING INFORMATION
TABLE 3–2

MAGNET	ALLOWABLE SHIPPING MODES	SHIPPING CAPABILITY	MAXIMUM TRANSIT TIME	COMMENTS
LCC 1.0T and 1.5T	Airplane, Air ride only trailer, Boat or ocean going ship	Warm or Cold	Cryogen refill for cold shipment = 15 days	See Note for Tie downs

3–2 IN–TRANSIT HANDLING AND SERVICE (continued)

MAXIMUM LOAD INFORMATION
TABLE 3–3

MAGNET	MAXIMUM LONGITUDINAL LOAD	MAXIMUM LATERAL LOAD	MAXIMUM VERTICAL LOAD	MAXIMUM SHOCK LOADS
LCC 1.0T and 1.5T	1.2 G	1.5 G	1.5 G	None Allowed

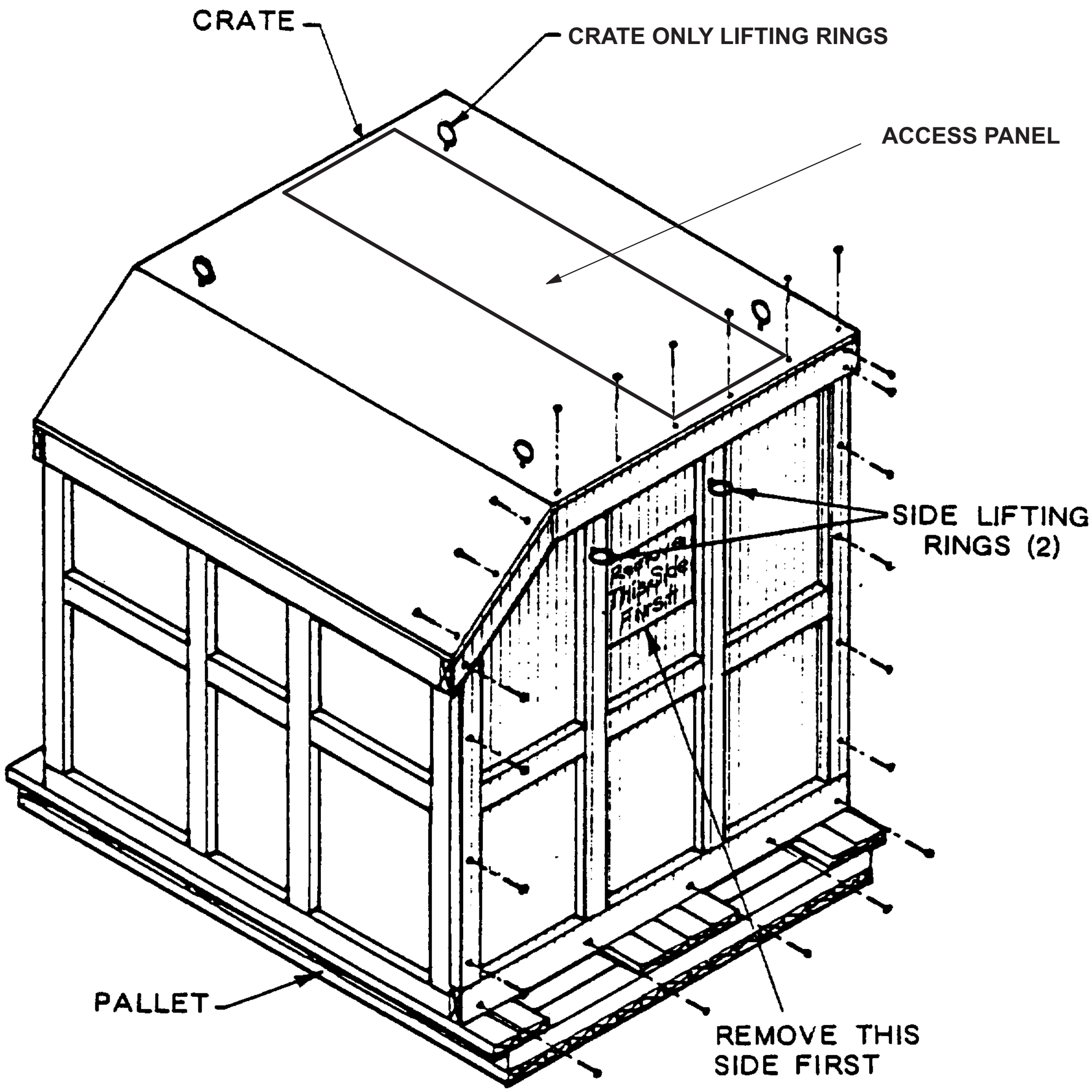
* Includes weight of BRM Coil – 2,300 lbs. (1,044 kg.) and associated hardware – 126 lbs. (58 kg.) including cryogens. Does NOT include Shipping Crate / Pallet – 2,200 lbs. (998 kg.)

** Maximum tilt allowed only when supported by feet for forklift / rigging operations or moving on inclines (base lift).
Tilt allowed when lifted by single ended slings is one deg. (top lift).

3–3 CRATE REMOVAL

1. Remove all subsystem crates, except the magnet crate, from low–boy trailer using a crane or forklift. Inspect all crates for visible damage.
2. Move subsystem crates to a receiving location protected from the weather, preferably in proximity of the examination room and on the same level.
3. Inspect the crate containing the magnet and identify the crate side marked, “REMOVE THIS SIDE FIRST”. This designated side is to be removed first.
4. Position crane centrally over this designated crate side.
5. Connect crane to the two lifting rings located near the top of the crate side marked “REMOVE THIS SIDE FIRST” using a sling and shackles supplied by riggers. Tighten the sling snug. See Illustration 3–x.
6. Remove lag screws securing the face of the crate marked “REMOVE THIS SIDE FIRST” See Illustration 3–1.
7. Lift crate side with crane and clear from the area.

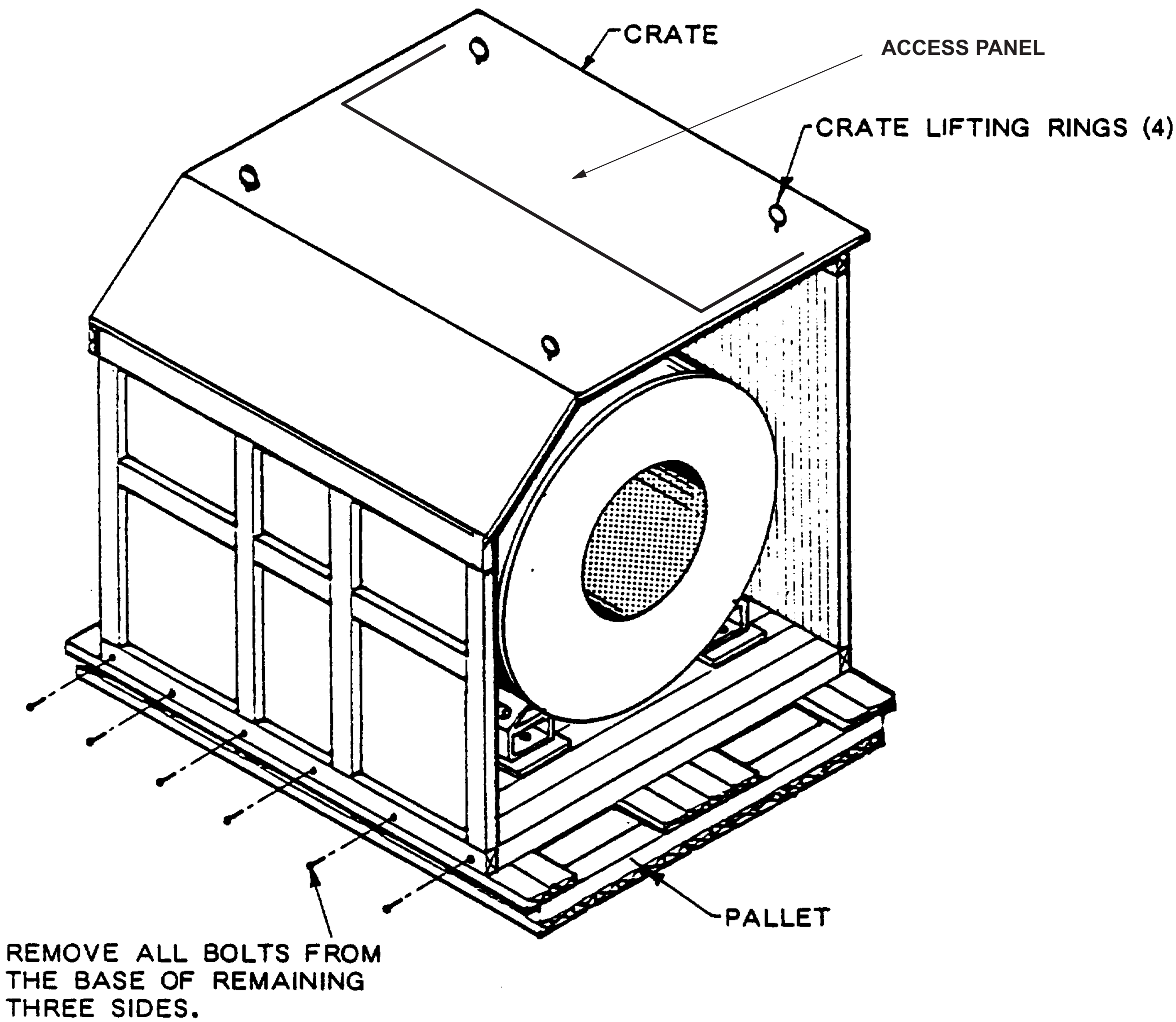
3–3 CRATE REMOVAL (continued)



MAGNET CRATE FACE LAG SCREW REMOVAL
ILLUSTRATION 3–3

3–3 CRATE REMOVAL (continued)

8. Secure four one ton (907 Kg) working load slings, minimum 6 foot length (1829mm), to four lifting rings assembled to the top corners of crate. Using one ton (907 Kg) anchor shackles.



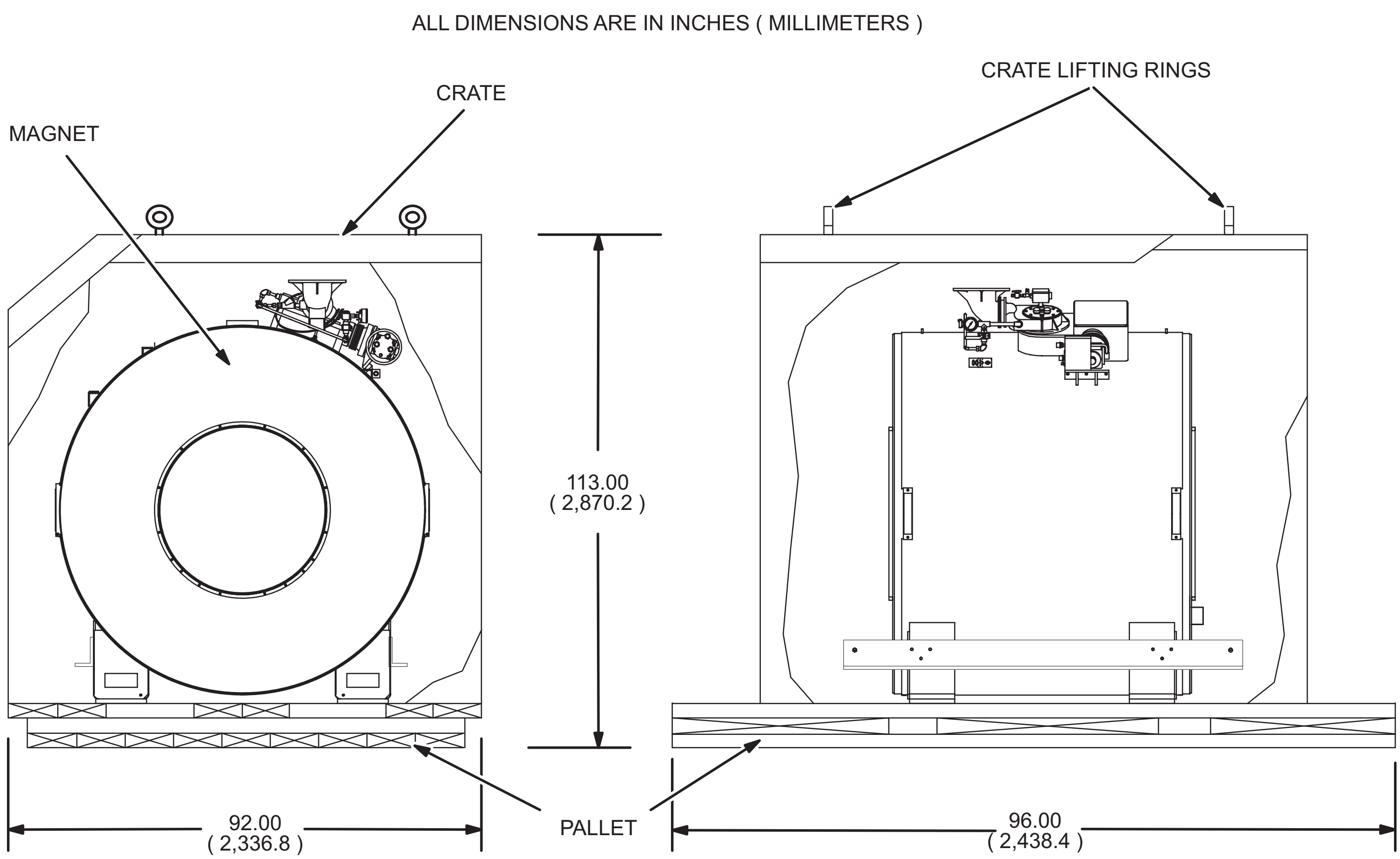
MAGNET CRATE BOTTOM LAG SCREW REMOVAL
ILLUSTRATION 3–4

3–3 CRATE REMOVAL (continued)

9. Position crane centrally over top section of the crate and attach slings to crane hook.
10. Remove lag screws only from the bottom portions of the three remaining crate sides. See Illustration 3–4.
11. Lift crate carefully using crane and clear crate from shipping pallet containing magnet.

CAUTION

Make sure that the crate does not swing and hit the magnet coldhead, vertical penetration or plumbing during the lifting process.



MAGNET CRATE
ILLUSTRATION 3–5

3–4 UNLOADING MAGNET AND PALLET

WARNING!

DO NOT APPLY ANY FORCE IN SHADED AREAS. A MAXIMUM LOAD OF 350LB. (PERSON, TOOLS, ETC.) IS ALLOWED ON TOP OF VESSEL DURING MAGNET SERVICE. SERIOUS MAGNET DAMAGE COULD OCCUR IF LOAD LIMITS ARE EXCEEDED.

USE LIFTING STRAPS AS SHOWN TO LIFT MAGNET. DO NOT USE CHAINS OR OTHER SUBSTITUTES FOR STRAPS AS DAMAGE TO VESSEL COULD OCCUR.

READ DELIVERY / INSTALLATION MANUAL (2192625) BEFORE MOVING MAGNET.

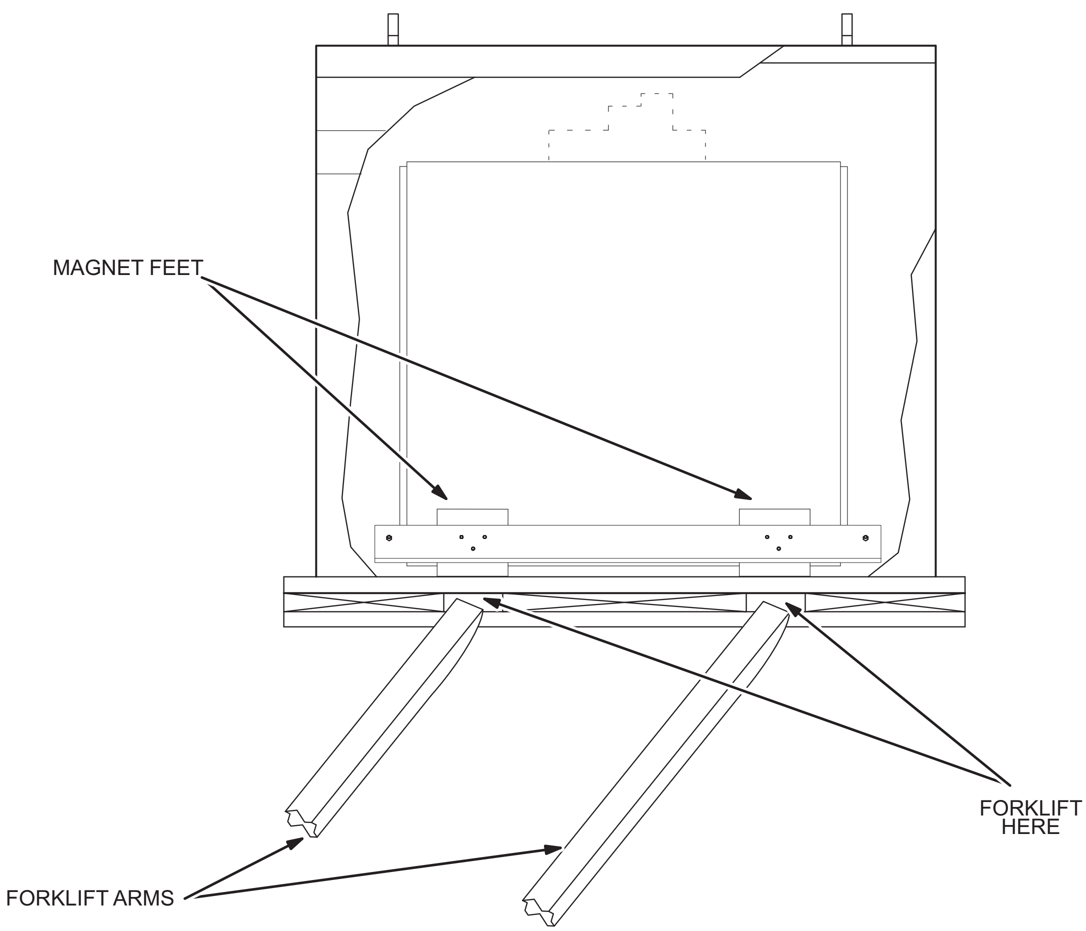
MAGNET NO–LOAD LOCATIONS
ILLUSTRATION 3–6

3–4–1 Forklift Method

CAUTION

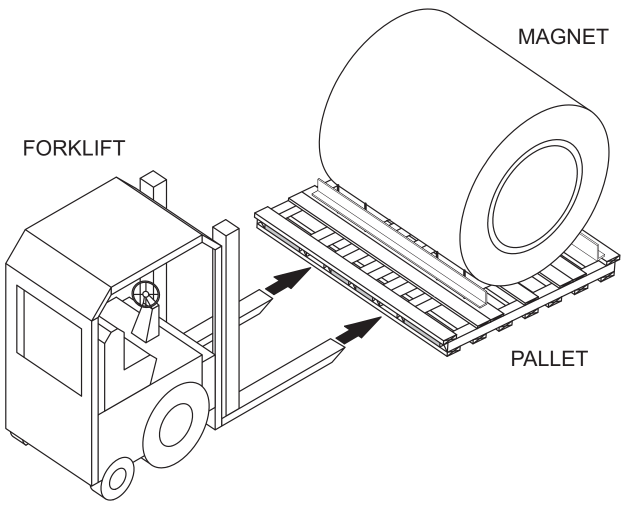
It is important to use the proper size forklift and to lift the magnet smoothly to avoid impact or jolts to the system which may cause damage to the magnet. Extreme care must be exercised during forklift operations. The magnet pallet must be picked up from the sides only. The forks MUST be placed under all four (4) feet of the magnet. Make sure the forklift capacity is rated for the weight of the crate and the contents. See Illustration 2–5 in Section 2 for required fork dimensions. The minimum fork length is 2.1 Meters (80 inches).

1. Locate the forklift openings on the side of the shipping pallet. See Illustration 3–7.
2. Extreme care must be exercised during forklift operations. The magnet / pallet must be picked up from the sides only. The forks MUST be placed directly under the four (4) feet of the magnet. Position the forks at the shipping pallet forklift openings. See Illustration 3–7.
3. Carefully drive the forklift until forks are completely inserted into the pallet openings.
4. Lift magnet / pallet with forklift and move smoothly to desired location.

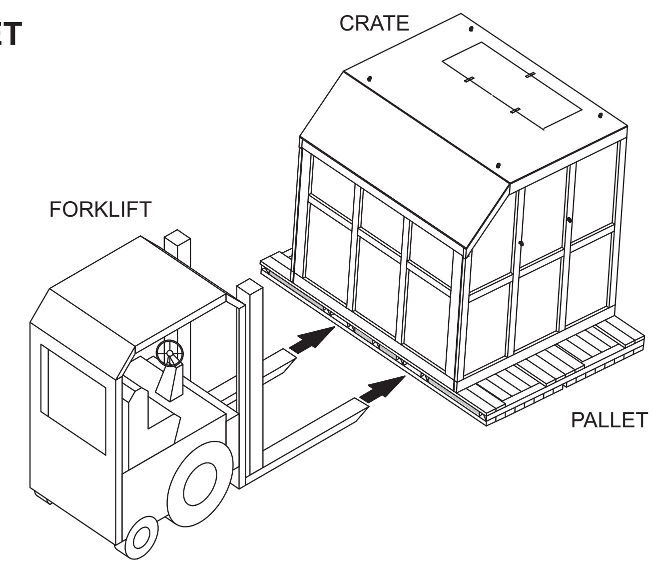


FORKLIFT / CRANE LIFTING OPERATIONS
ILLUSTRATION 3–7

3–4–1 Forklift Method (continued)



MAGNET AND PALLET



MAGNET, PALLET AND CRATE

LIFTING CONFIGURATIONS FOR FORKLIFT
ILLUSTRATION 3–8

3–4–2 Crane Method

CAUTION

The lifting hooks of the spreader beam MUST be located at the dimensions shown in Illustration 3–11. This dimension is based on 3.00" (76.2 mm) wide lifting straps or cables rated at 11,000 lbs. (4,990 kg.) minumum.

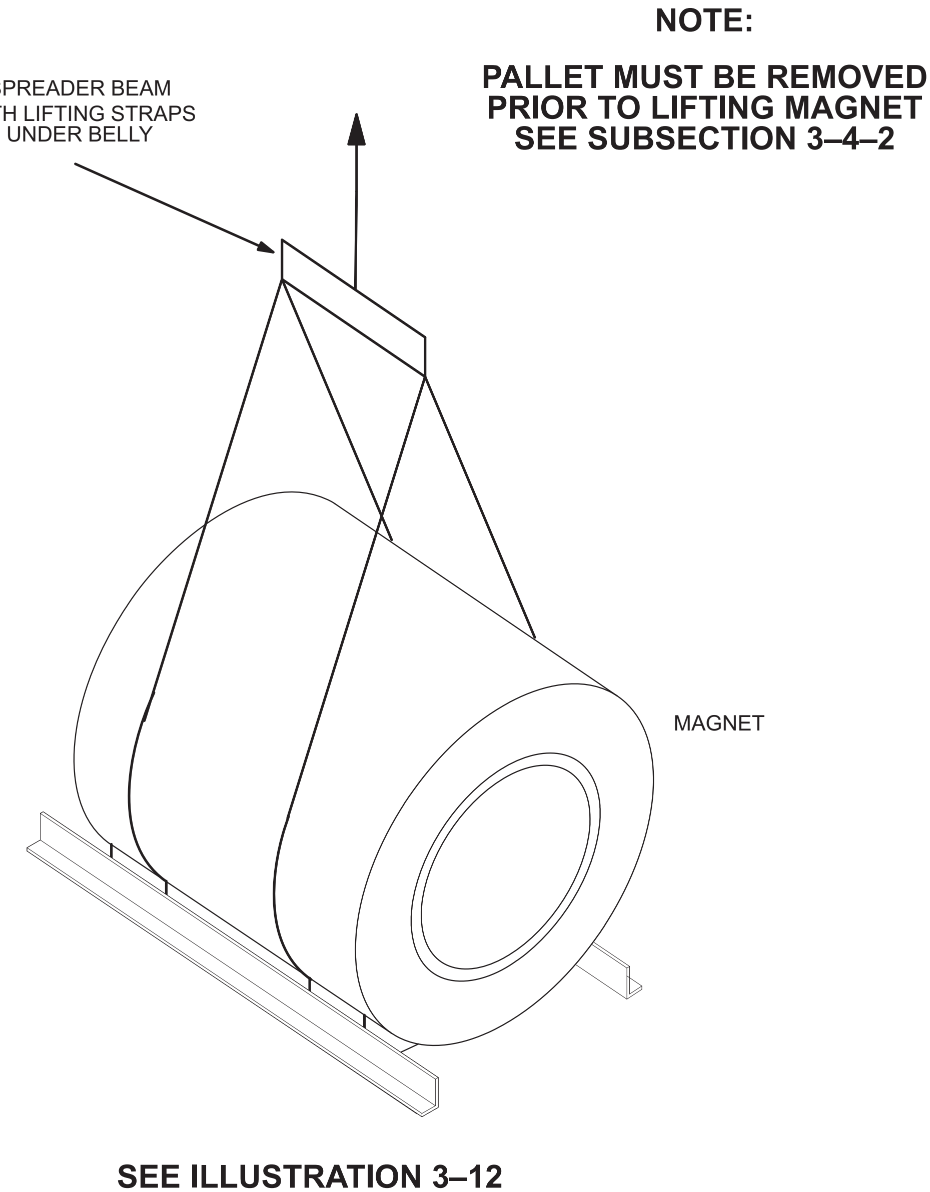
Lifting straps / cables must be long enough to allow a minimum of 24.00" (609.6 mm) clearance between the top of the magnet outer vacuum vessel and the top of the lifting strap / cable. See Illustration 3–11.

Shipping pallet CANNOT be lifted by magnet. Pallet MUST be removed from magnet prior to any lift operations.

Failure to follow these requirements can result in serious damage to the magnet.

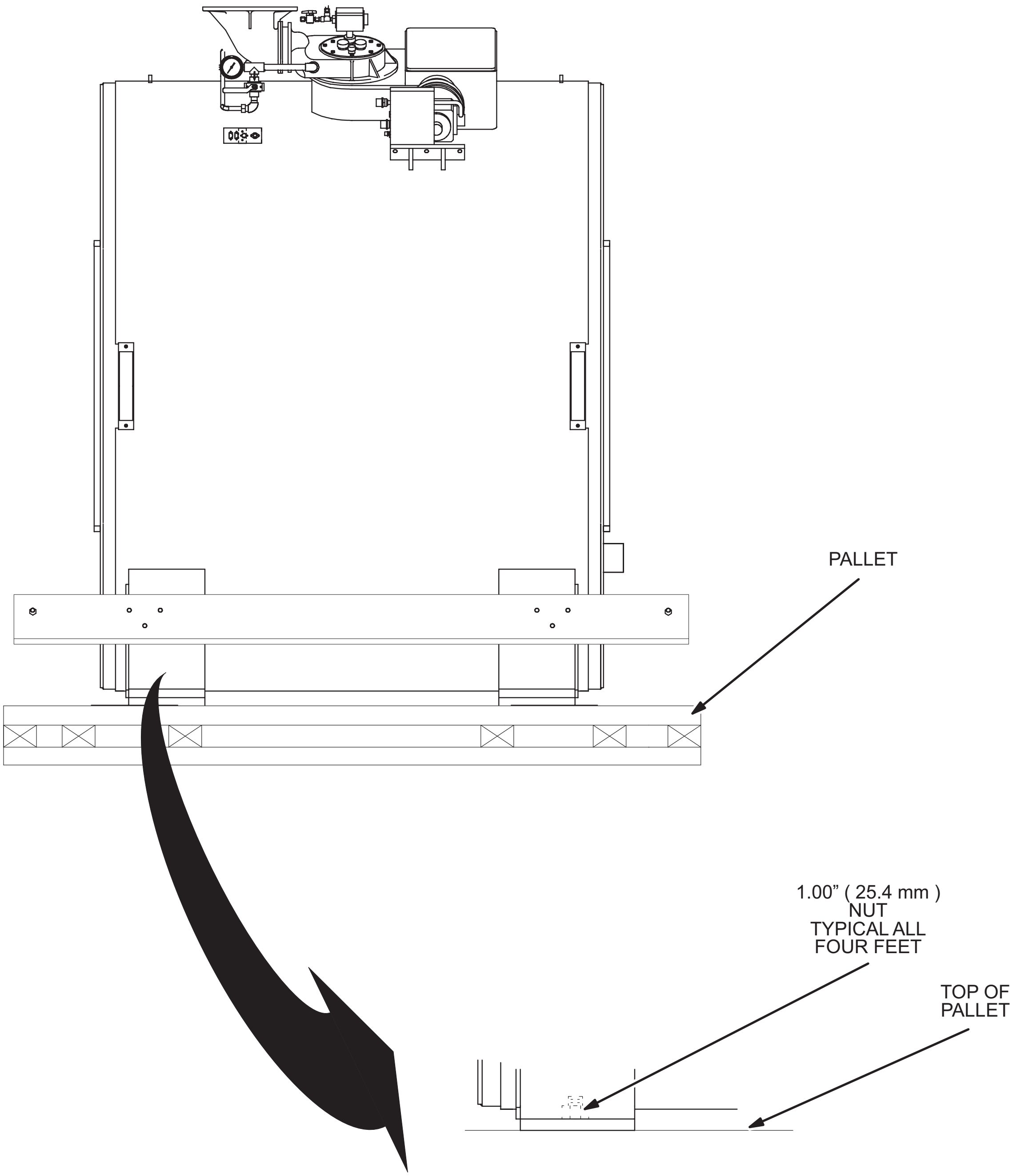
1. Unbolt and remove the four 1 inch (25.4 mm) nuts which secure the pallet to the magnet. See Illustration 3–10.

3–4–2 Crane Method (continued)



LIFTING CONFIGURATIONS FOR CRANE (UNDER BELLY)
ILLUSTRATION 3–9

3–4–2 Crane Method (continued)



MAGNET MOUNTING ON SHIPPING PALLET
ILLUSTRATION 3–10

3–4–2 Crane Method (continued)

CAUTION

The magnet must be moved as smoothly as possible at all times. If severe jolting is encountered, internal damage may result.

2. Position the crane hook centrally over the magnet to ensure a vertical lift force on the lifting straps / cables. See Illustrations 3–11 and 3–12.

CAUTION

It is important to lift the magnet smoothly to avoid impact or jolts to the system which may damage the magnet. The magnet cannot be tipped by more than 1 degree during crane lifting operation.

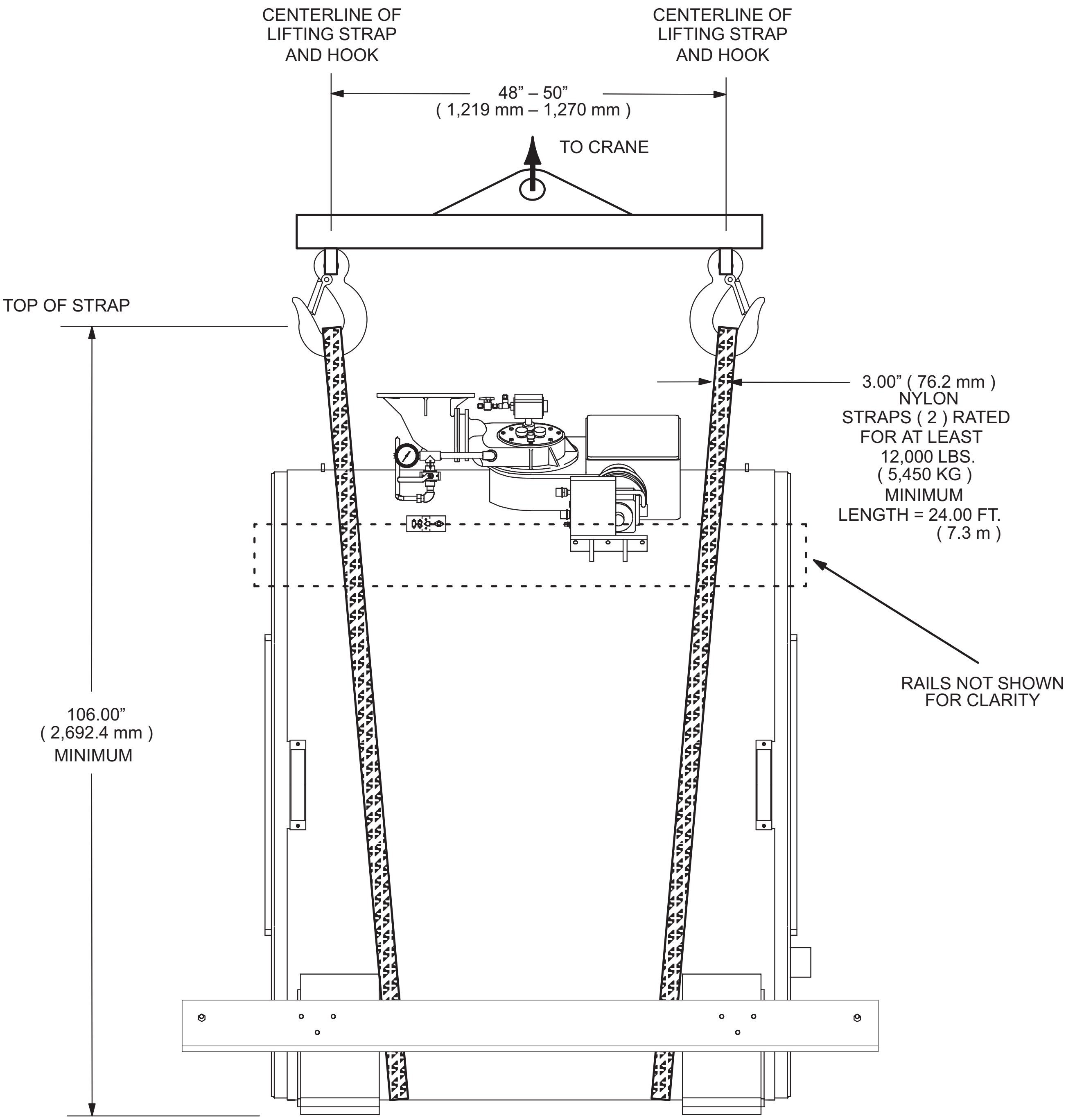
3. Attach the rigging to the lifting straps / cables at each end of the magnet. See Illustrations 3–11 and 3–12.
4. Lift magnet with crane to clear shipping pallet.

CAUTION

When lowering the magnet, be careful not to shock the magnet which could result in over stressing the internal magnet support members.

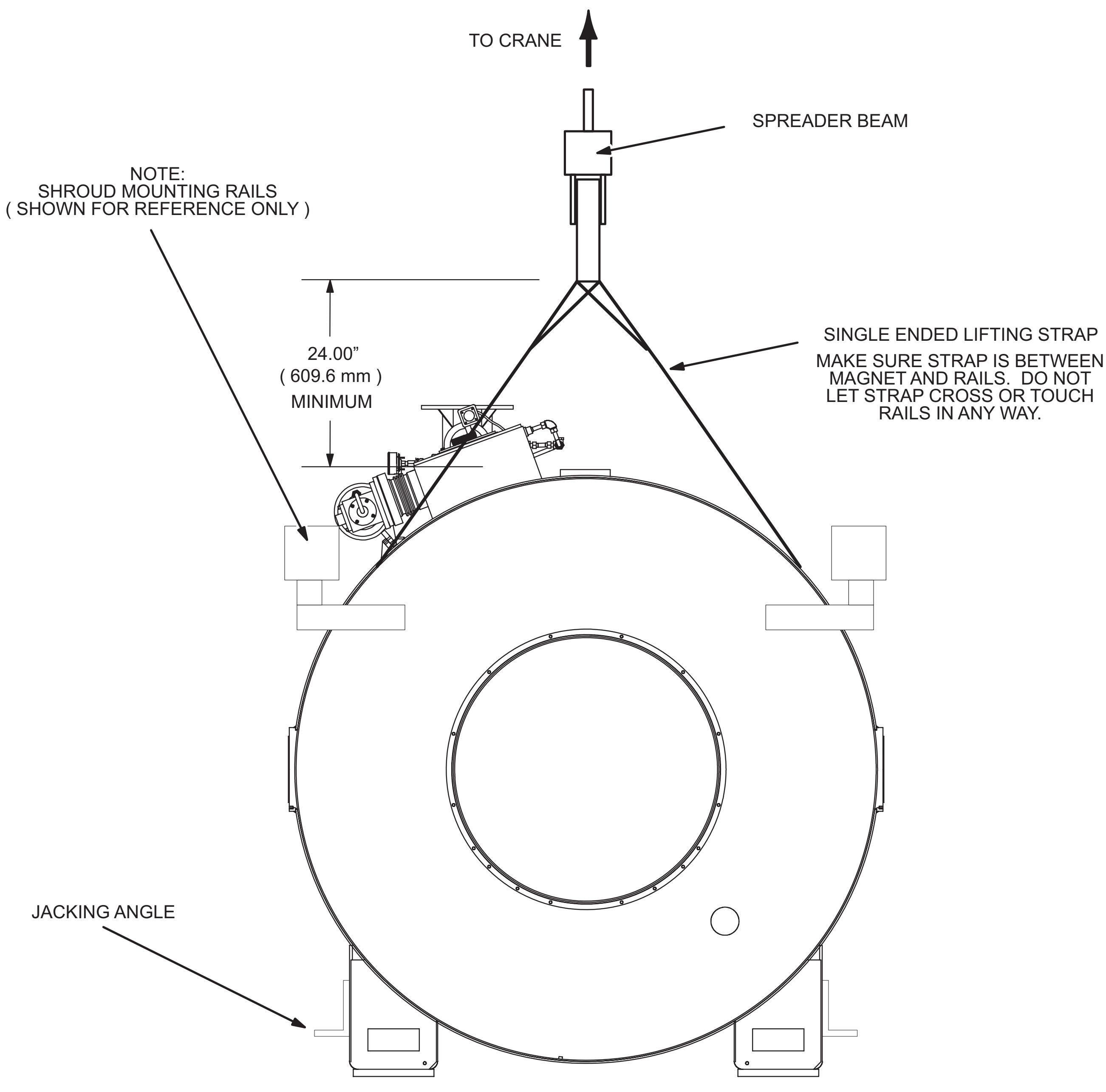
5. Lower the magnet onto a flat surface.

3–4–2 Crane Method (continued)



MAGNET LIFTING STRAP LOCATIONS
ILLUSTRATION 3–11

3–4–2 Crane Method (continued)



MAGNET LIFTING STRAP HEIGHT
ILLUSTRATION 3–12